

# Problem Set 2

Tate Mason

November 9, 2025

## Problem 1: Data Overview and Analysis

### Part A: Data Summary

In Colombia, we see fewer overall observations in every sector:

- Food Industry: 6140
- Apparel Industry: 4454
- Metal Industry: 3678
- Textile Industry: 2840
- Wood Industry: 964

So, we see that Colombia has a huge food and manufacturing sector, whereas they have a smaller industry producing raw materials like wood.

In Chile, we see an interesting distribution of observations:

- Food Industry: 18833
- Metal Industry: 5856
- Textile Industry: 5447
- Wood Industry: 4835
- Apparel Industry: 4694

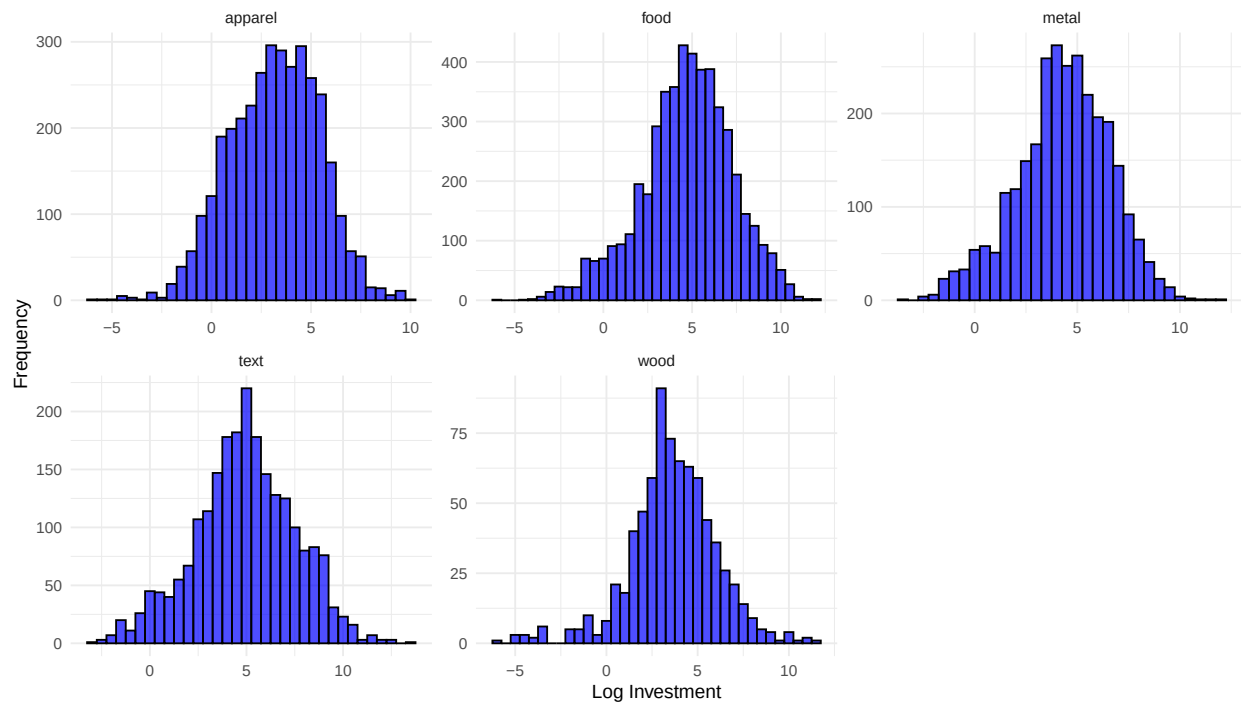
So, Chile has a massive food industry, but also has a balanced manufacturing sector with a lot of metal, textile, and wood production.

Over time, we see varying trends, indicating some industrial growth and some decline in different sectors for both countries. Further, what may be a data aggregation issue, 1981 has disproportionately more observations than other years for both countries. Something to look back at.

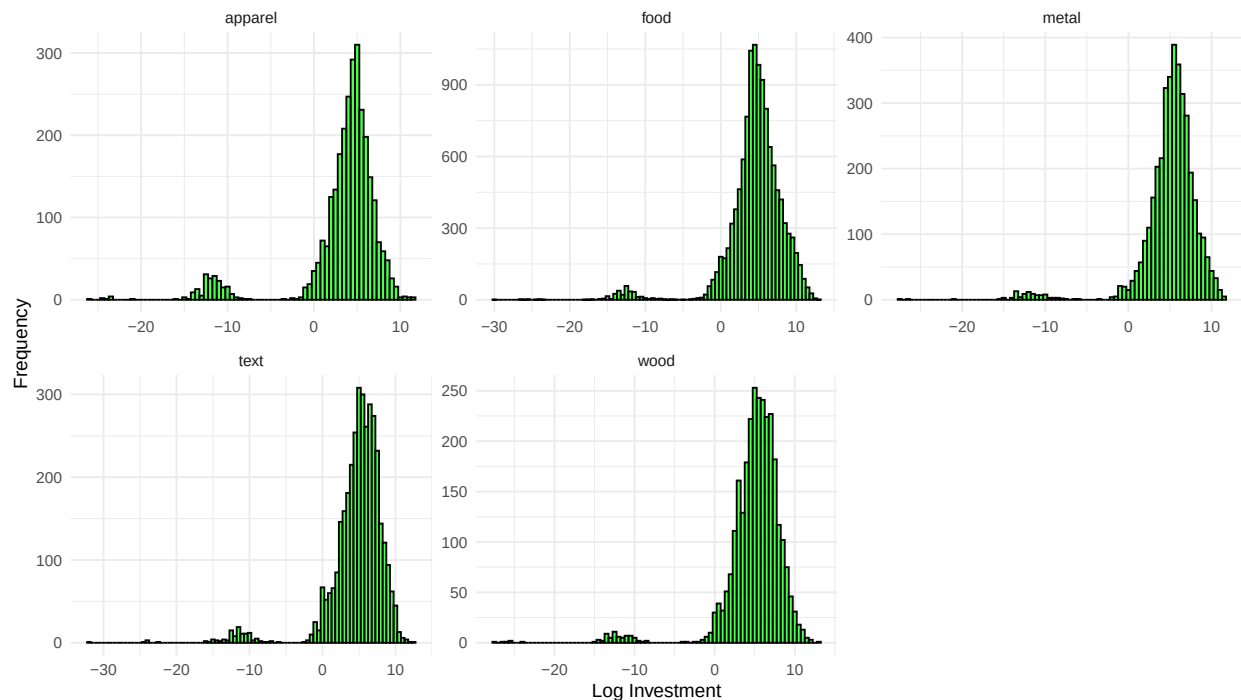
| Year | Colombia Observations | Chile Observations |
|------|-----------------------|--------------------|
| 1979 |                       |                    |

## Part B: Analysis of Investment by Industry

Histogram of Log Investment by Industry, Colombia



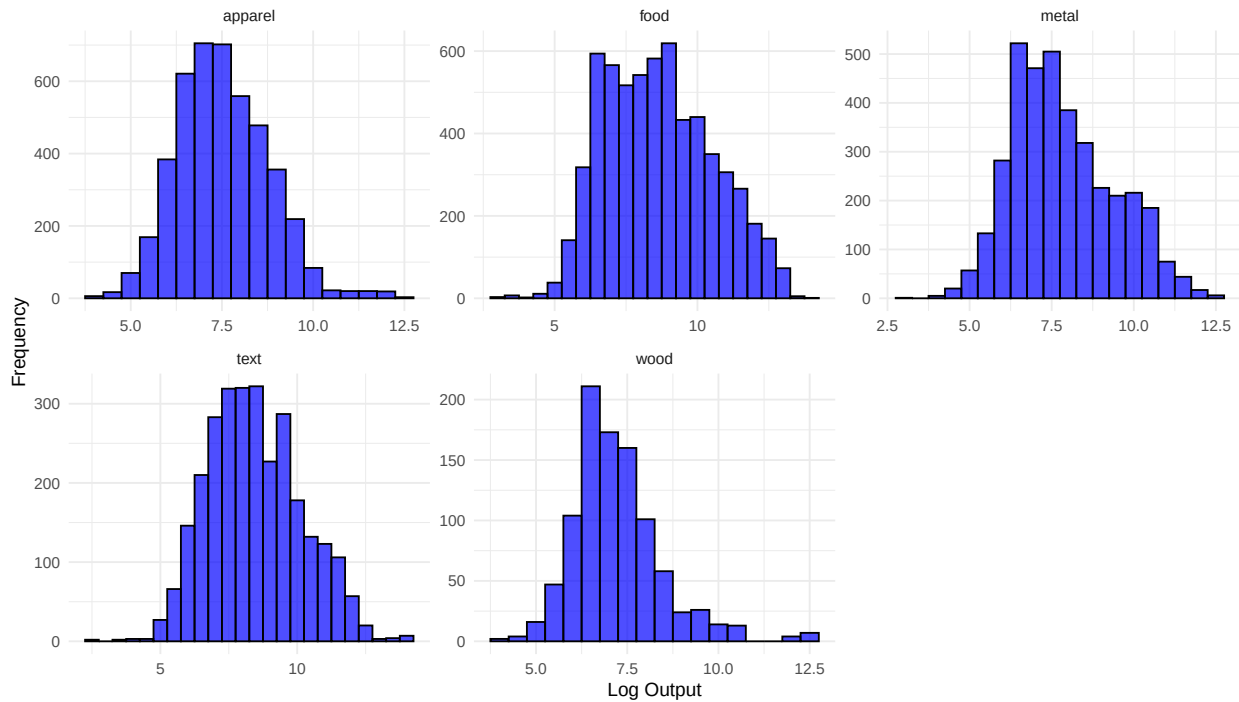
Histogram of Log Investment by Industry, Chile



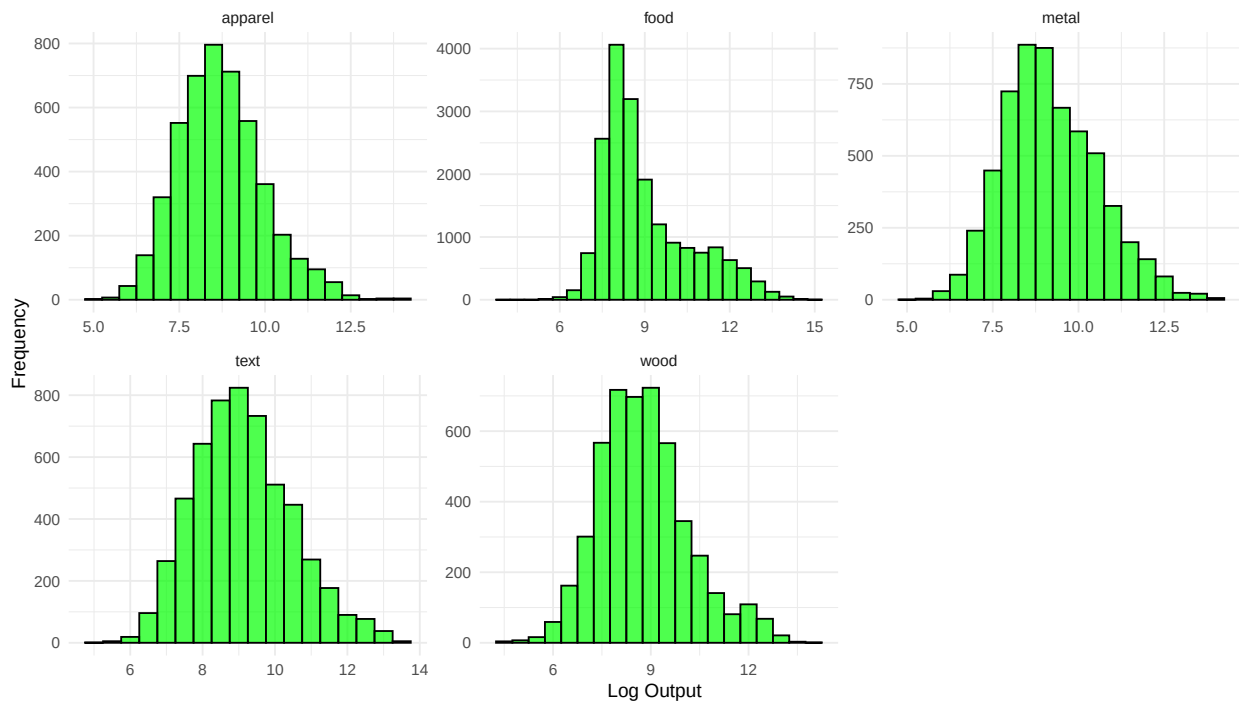
In Colombia, there is much more dispersion in investment levels across industries, with some industries even experiencing negative investment. Food, metal, and textiles have very similar investment patterns, with means around 5. Apparel and wood are similar to the other three, though a bit lower. In Chile, investment levels are super consistent across industries, with all industries having means around 5-7. Distributions are also much tighter, though most do have some negative investment observations.

## Part C: Analysis of Ouput by Industry

Histogram of Log Output by Industry, Colombia



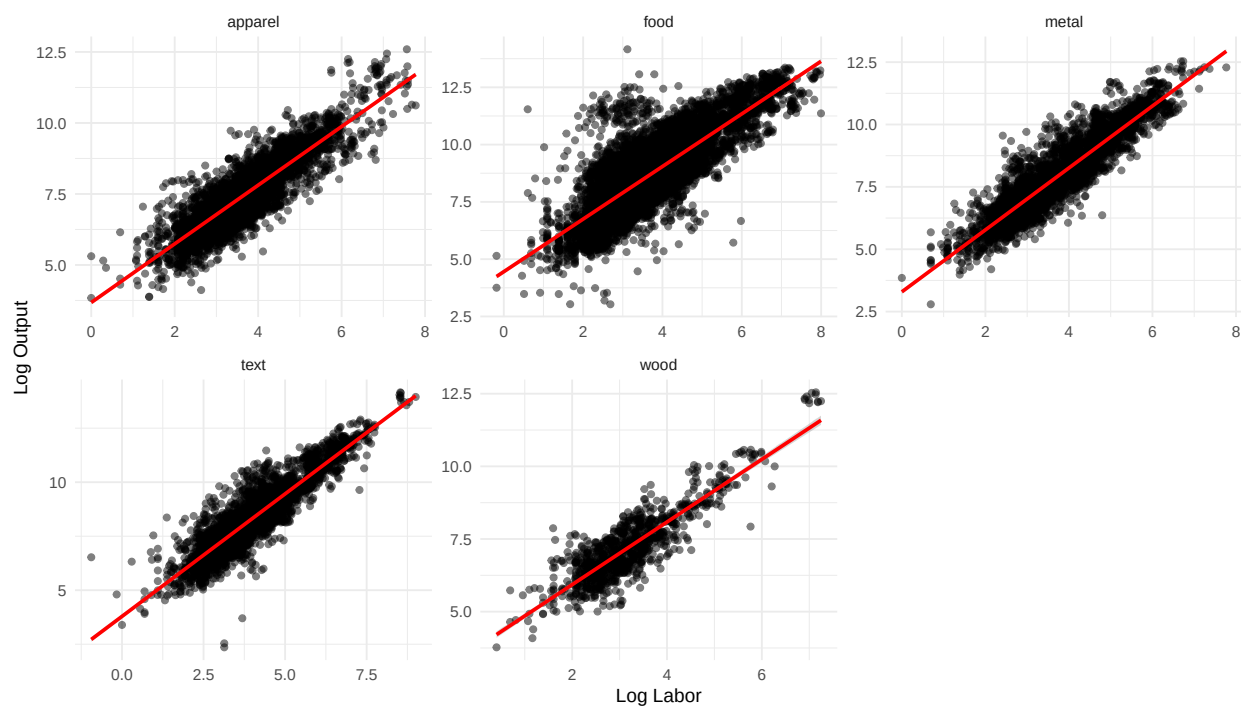
Histogram of Log Output by Industry, Chile



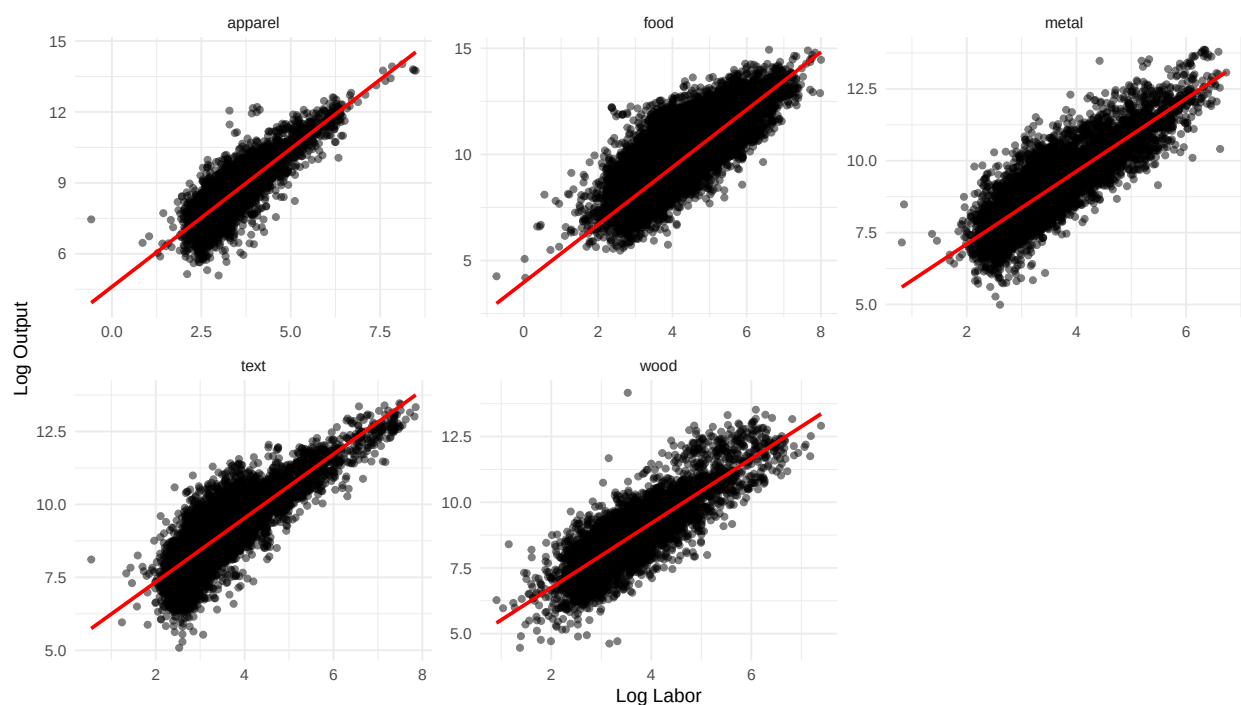
In Colombia, output levels do not vary too much across industries, with means around 7-10 and similar distributions. In Chile, distributions are also similar, though there is a bit more variation. Textiles and food have a wider distribution of output levels, while apparel and metal are a bit tighter.

## Part D: Analysis of Labor Output Correlation by Industry

Scatter Plot of Log Output vs Log Labor (Colombia)



Scatter Plot of Log Output vs Log Labor (Chile)



In both countries, there is a positive relationship between labor input and output across all industries. The strength of this relationship varies by industry, with some industries showing a stronger correlation than others. For example, in Colombia, the food industry shows a strong positive correlation, while the wood industry has a weaker correlation. In Chile, all industries show a strong positive correlation.

## Problem 2: Estimation Techniques

Part A: OLS

Part B: OLS with Investment

Part C: OP Estimation

Part D: LP Estimation

Part E: ACF Estimation

## Problem 3: Productivity Dynamics

Part A: Estimation of  $\omega$  Using OP Estimates - Homogenous

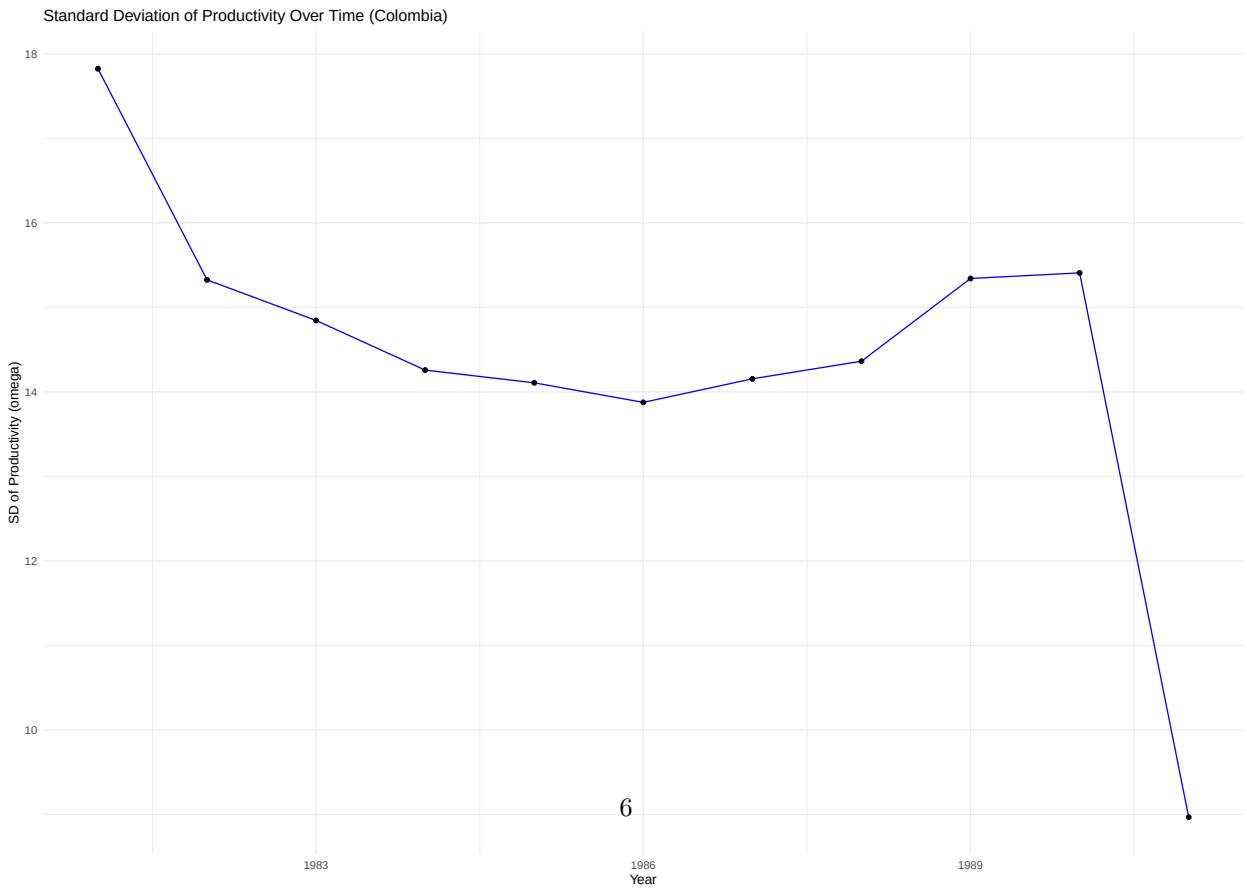
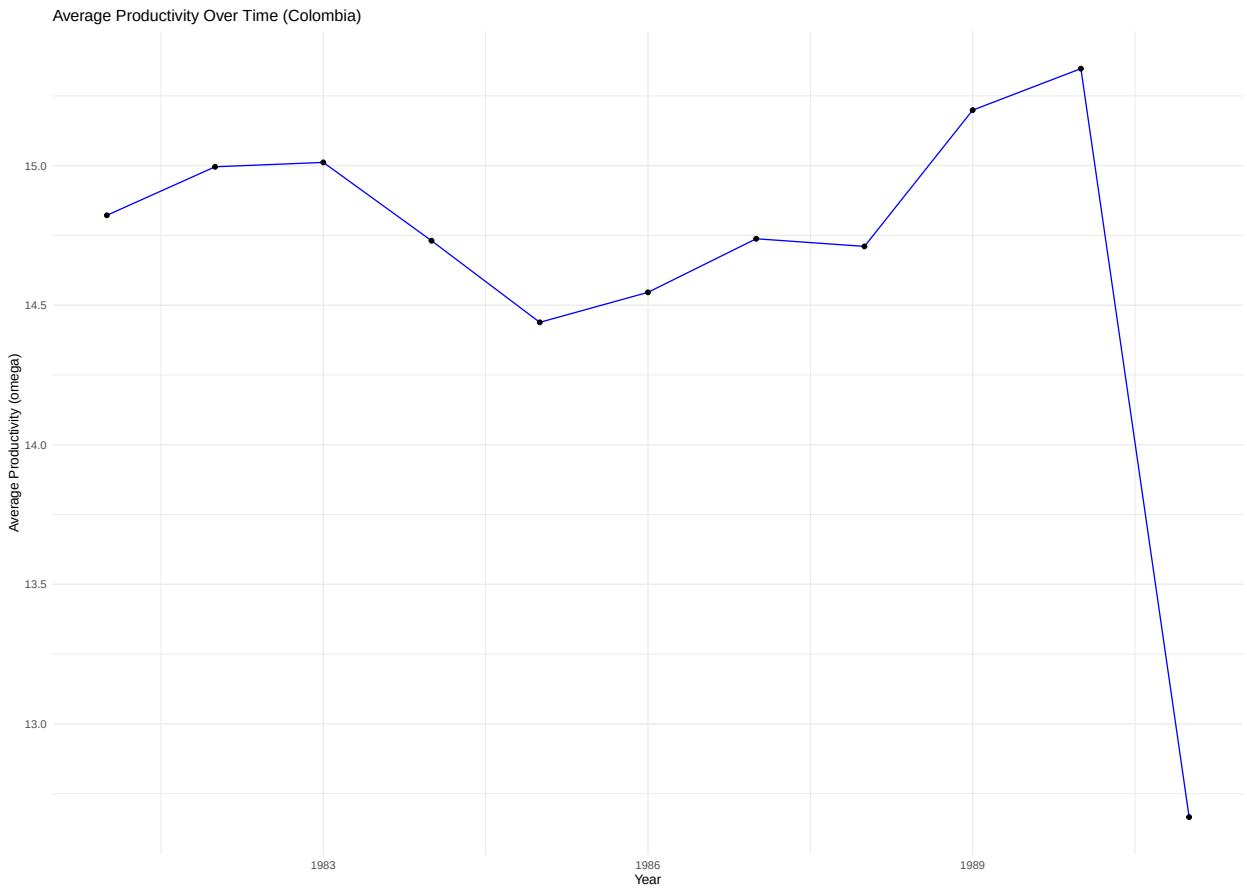
| Country  | Mean of $\omega$ | Std. Dev. of $\omega$ |
|----------|------------------|-----------------------|
| Colombia | 110.323          | 310.548               |
| Chile    | 81.732           | 153.754               |

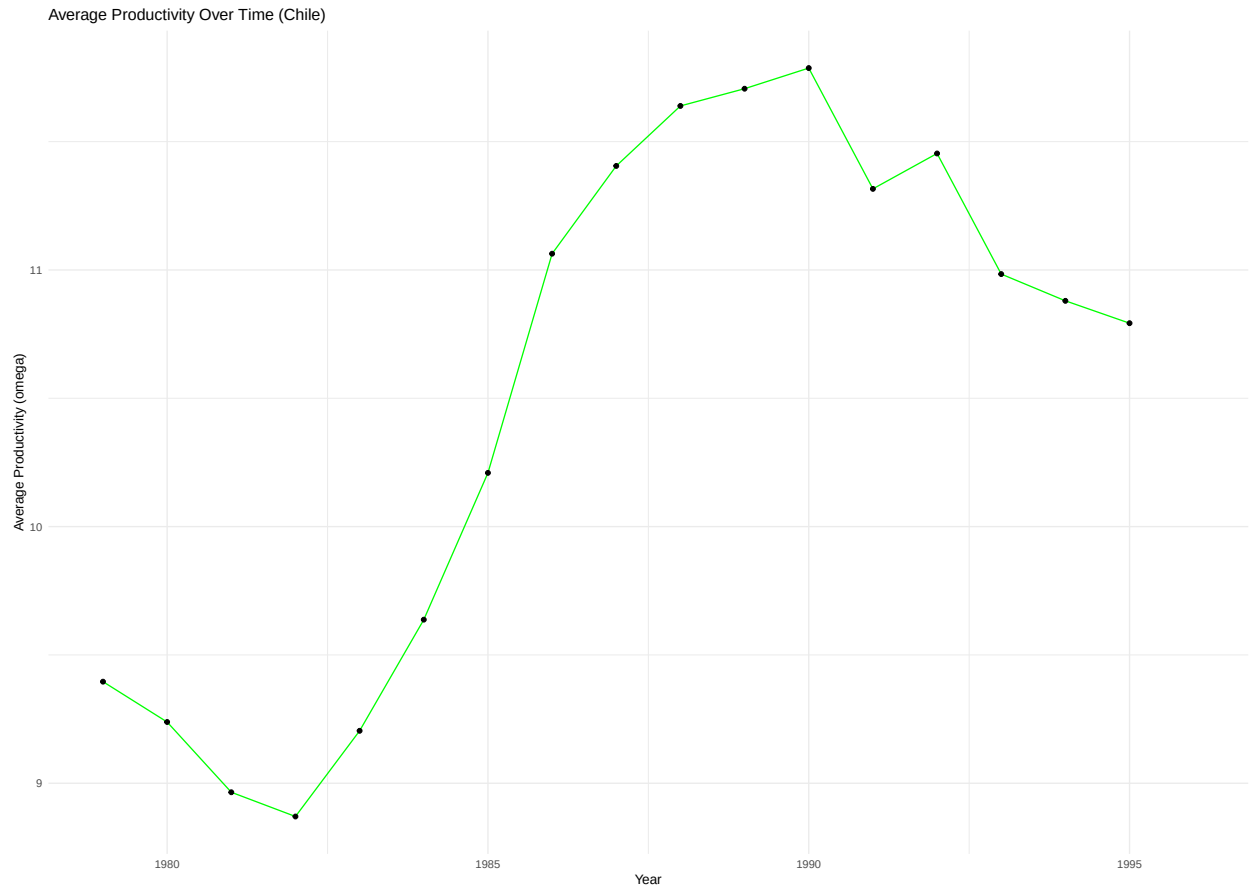
Part B: Estimation of  $\omega$  Using OP Estimates - Heterogenous

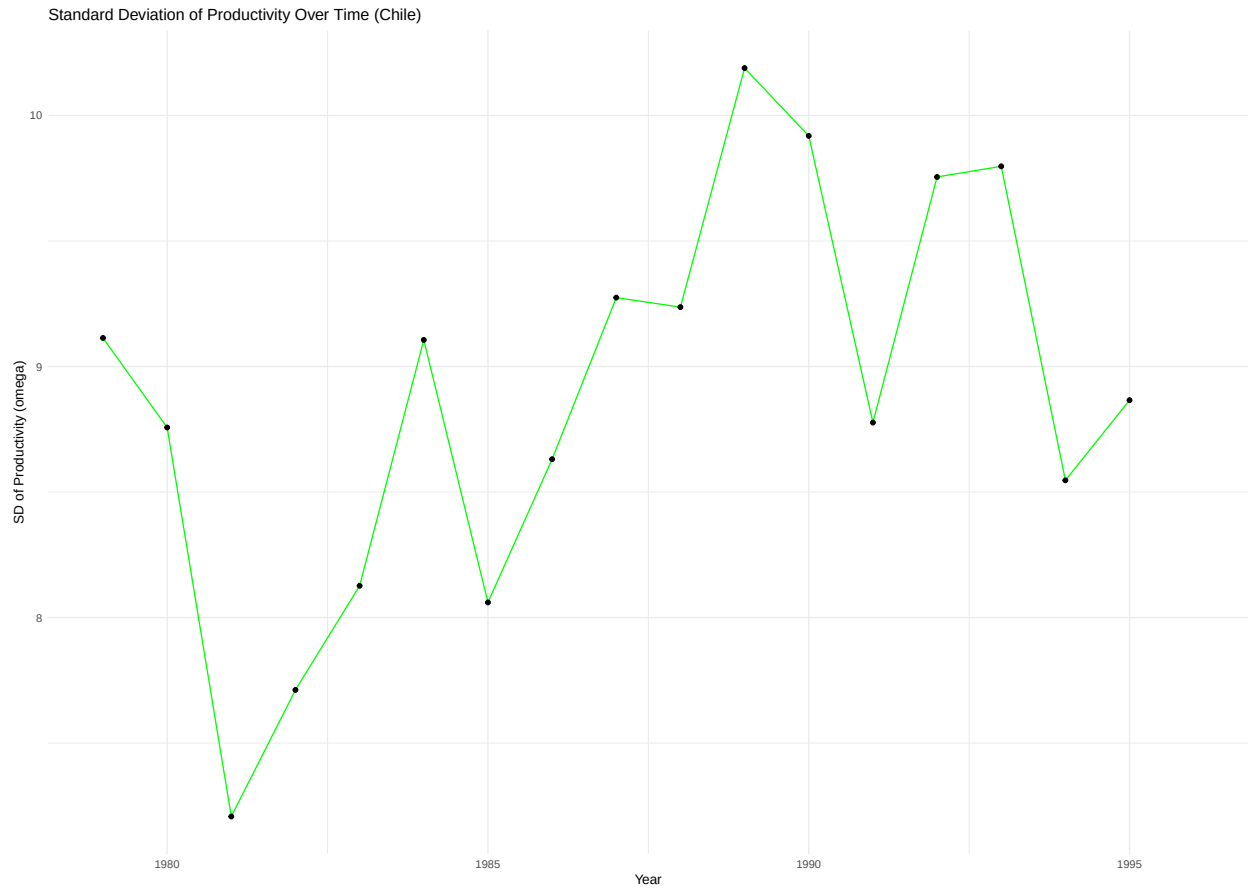
| Country  | Industry | Mean of $\omega$ | Std. Dev. of $\omega$ |
|----------|----------|------------------|-----------------------|
| Colombia | Food     | 110              | 160                   |
|          | Apparel  | 111              | 92.8                  |
|          | Metal    | 100              | 69.2                  |
|          | Textile  | 159              | 125                   |
|          | Wood     | 341              | 370                   |
| Chile    | Food     | 33.2             | 32                    |
|          | Apparel  | 23.3             | 17.7                  |
|          | Metal    | 44.3             | 30.6                  |
|          | Textile  | 43.4             | 43.2                  |
|          | Wood     | 54.1             | 43.9                  |

# Part C: Time Trends in Productivity

## Homogenous Productivity Trends

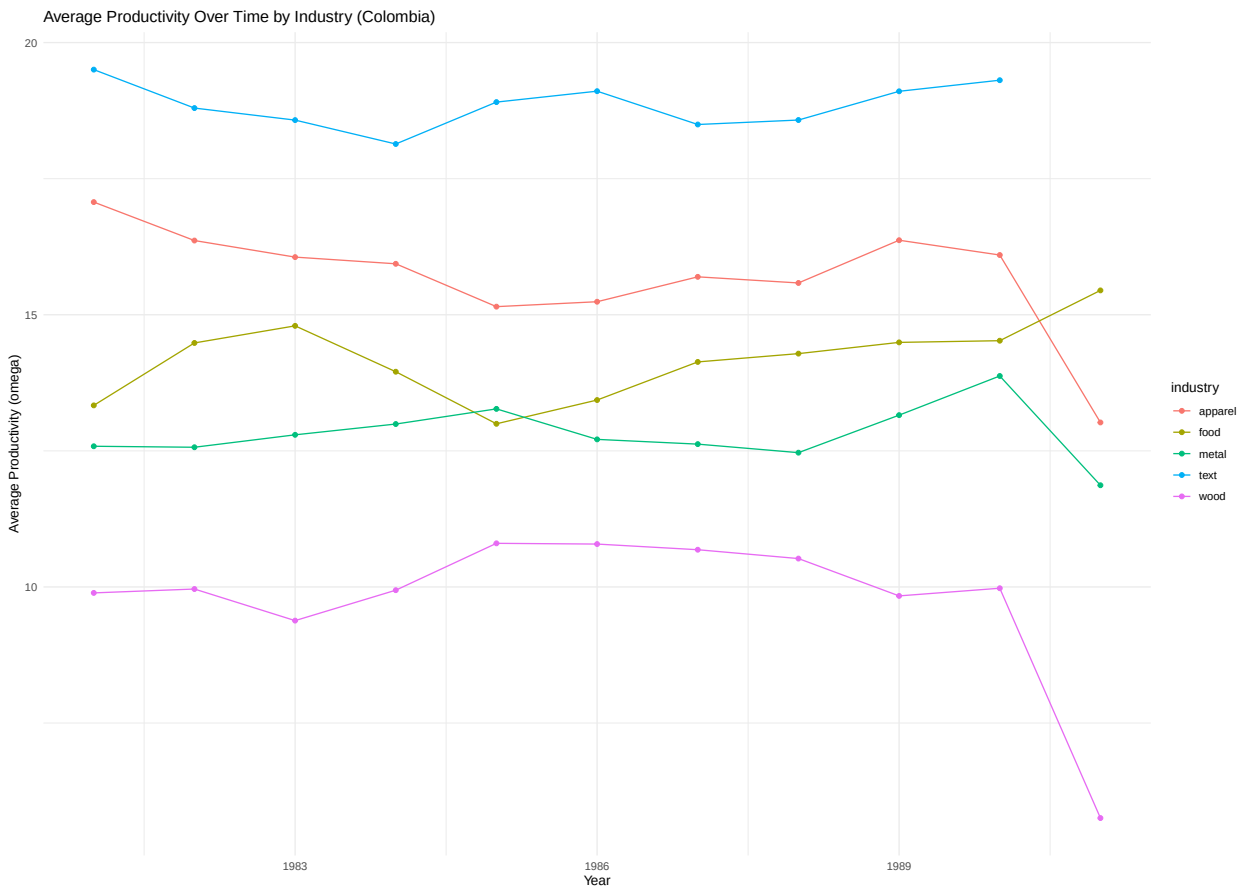


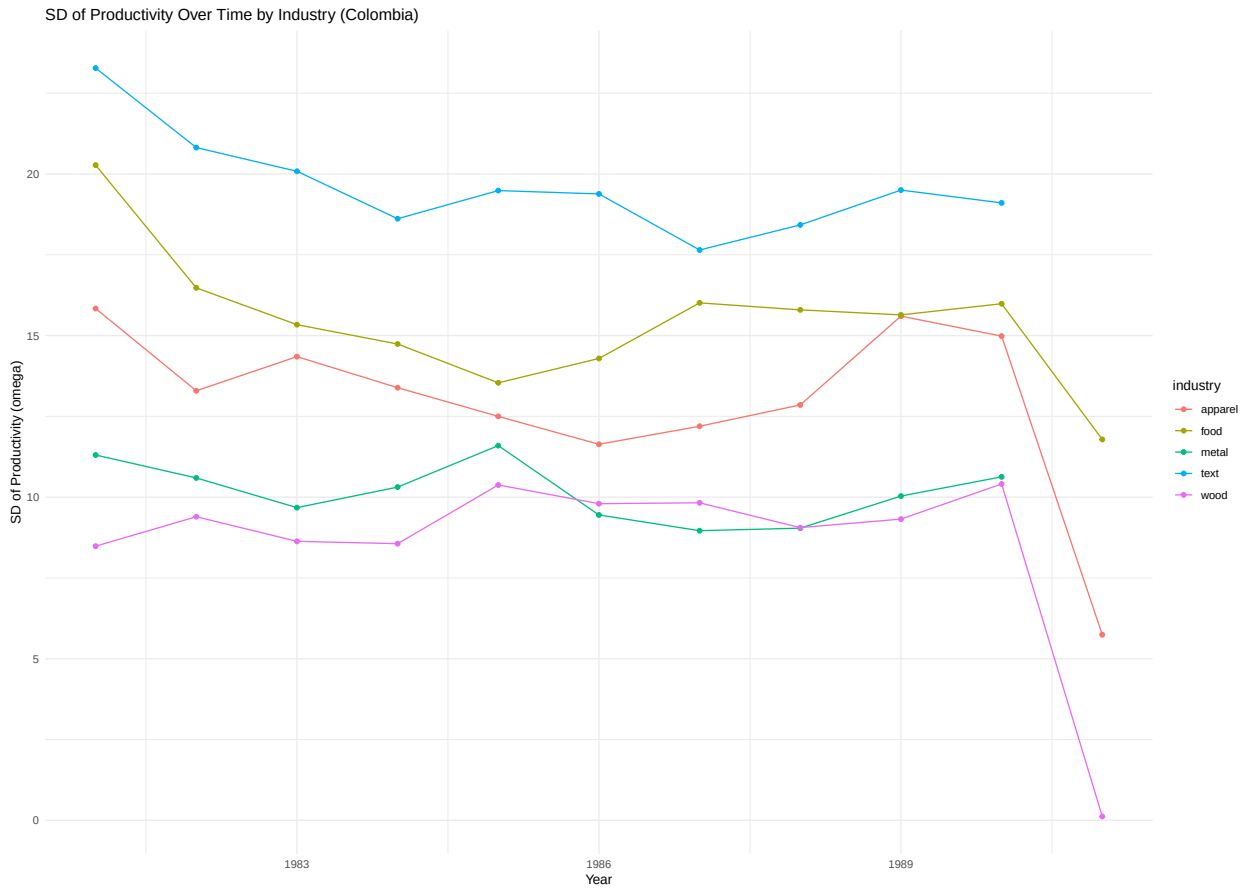


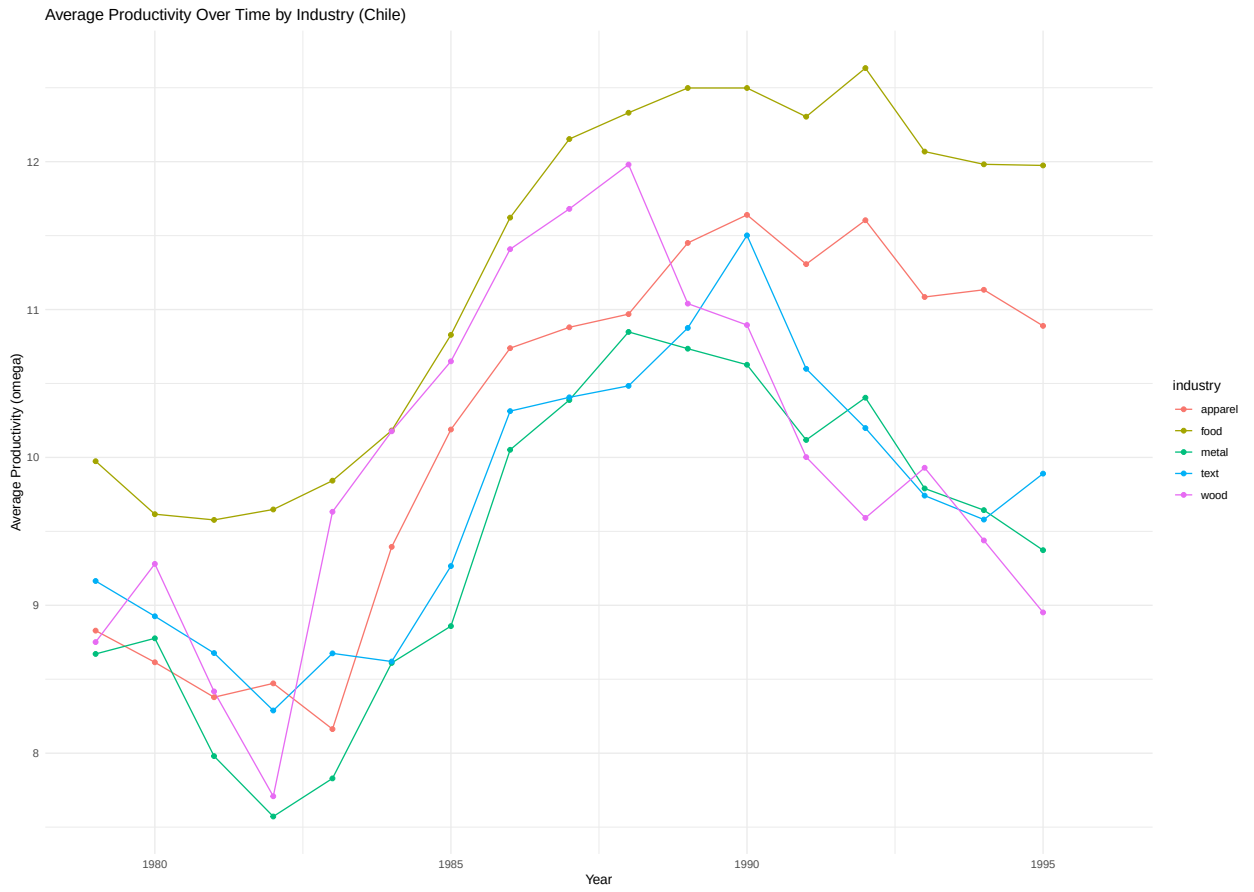


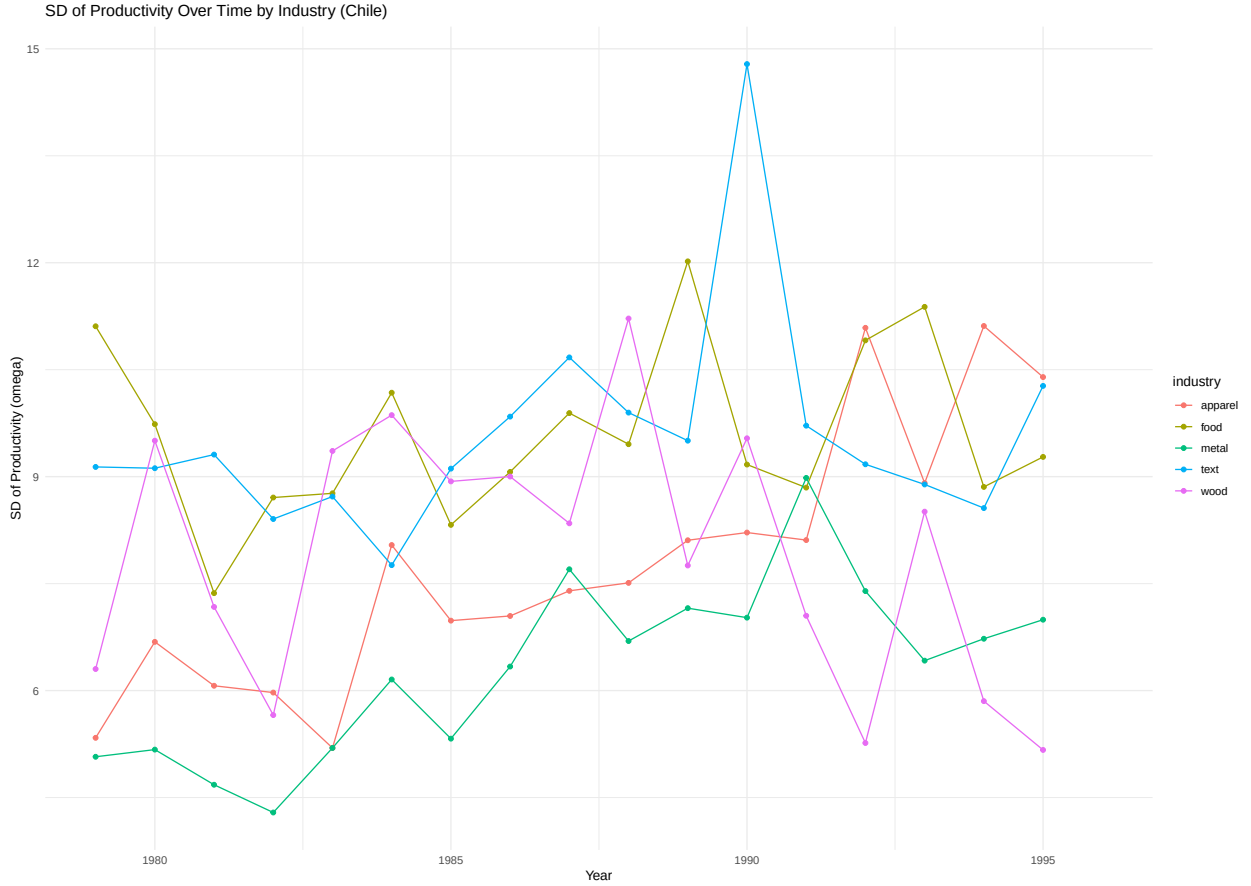


# Heterogenous Productivity Trends by Industry









## Discussion

In Colombia, average productivity trends upward over the time horizon, with a slightly lesser increase in standard deviation, indicating that the firms are becoming more productive on average with time, while variability is not increasing to the same degree. In Chile, we see a spike in productivity after 1985, continuing until 1990 before slowing. Standard deviation also increases over time, with more consistency in growth. This suggests that Chilean firms are also becoming more productive on average, but with greater variability among firms.

When splitting out by industry, we see that Colombian textile industries are the most productive on average, while wood and apparel are the least. However, the textile industry also exhibits the highest variability, while food is the least so. In Chile, all industries are increasing in productivity over time to similar degrees, with textiles being the most productive until about 1992 when food overtakes it. Apparel is the least productive until about 1994, when metal takes over as least productive. Food and apparel are the most variable industries in Chile when looking at growth in standard deviation, though textiles have the highest level until about 1990. Over the whole horizon, wood and metal are the least variable.